

Deterministic Extraction Architecture | Phase 1

No LLM in extract. Six specialist Python stages (pbixray). Optional later: NL Q&A agent on top of inventory.

INPUT

ROLE: Source

- .pbix semantic model only
- pbixray extraction (no row dumps)

ORCHESTRATOR

ROLE: Deterministic control

- Run stages 1-5 then Merger 6
- Never invent objects / no LLM

1 SCHEMA

ROLE: Structure

- Every table + column
- Types / calc flags
- LocalDate* internals

2 RELATIONSHIPS

ROLE: Graph

- From/to keys
- Cardinality
- Cross-filter / active

3 DAX

ROLE: Logic

- Full measure DAX
- Calc columns
- Calc tables

4 POWER QUERY M

ROLE: Prep

- Verbatim M
- 04_m_raw/*.m
- SQL/seed tags

5 TM EXTRAS

ROLE: Metadata

- Partitions
- Hierarchies / RLS
- Auto dates

6 MERGER

ROLE: Combine + summarize

- OBJECT_INVENTORY.md + OBJECT_COUNTS.json + printed summary counts
- Then generate DATA_MODEL.md / .pdf / .png

MAKE IT AGENTIC: NL conversation on top (future)

ROLE: Answer questions about the CURRENT PBIX only

- After each upload, inventory is replaced for that PBIX. Enable NL/LLM chat over that inventory.
- Users can ask any question about the updated model. Extract stays deterministic; Q&A is the agentic layer.

What each extract stage does

1 Schema

Deterministic structure extract: tables, columns, types, calc-table flags, internal auto-date tables.

2 Relationships

Deterministic join-graph extract: cardinality, cross-filter, active/inactive.

3 DAX

Deterministic logic extract: full measure / calc column / calc table expressions (no rewrite).

4 Power Query M

Deterministic prep extract: verbatim M to 04_m_raw + SQL/seed/union tags.

5 TM Extras

Deterministic metadata extract: partitions, hierarchies, RLS/OLS, annotations, auto dates.

6 Merger

Deterministic merge: OBJECT_INVENTORY.md + OBJECT_COUNTS.json + summary for UI/PDF.

Make it agentic: natural-language conversation on top

By enabling natural-language conversation on top of this deterministic extract, the system becomes agentic: users can ask any question about the PBIX that was just uploaded/updated. The agent reads only the current inventory (replaced each run). Extract itself never uses an LLM and never invents objects.